

Capability & Spatial Negotiation Algorithm

How `<picture>` constructs a multi-dimensional software evaluation tree using media and type attributes.

HTML Structure

```
<picture>

  <source media="(min-width: 800px)" type="image/avif"
    srcset="desktop.avif">

  <source media="(min-width: 800px)" type="image/webp"
    srcset="desktop.webp">

  

</picture>
```

Software Version Dependency

The inclusion of `type` inherently couples the layout to the **browser's software version**.

Older software engines (e.g., Safari 13, IE11) lack the decoders for modern formats like AVIF or WebP. When the parser evaluates the `type` attribute, it acts as a boolean capability check.

If capability check fails (False): The algorithm abandons that `<source>` node entirely and proceeds down the DOM tree, regardless of whether the spatial media query was true.

Browser Layout & Capability Evaluator (e.g., Blink/WebKit)

Top-to-Bottom Engine Evaluation Logic

